Osteoarthritis and Cartilage 18 (2010) 621–628

Osteoarthritis and Cartilage

OARSI OSTEOARTHRITIS RESEARCH SOCIETY INTERNATIONAL

Hip strengthening reduces symptoms but not knee load in people with medial knee osteoarthritis and varus malalignment: a randomised controlled trial

K.L. Bennell †*, M.A. Hunt †, T.V. Wrigley †, D.J. Hunter ‡, F.J. McManus †, P.W. Hodges §, L. Li ‡, R.S. Hinman †

† *Centre for Health, Exercise & Sports Medicine, School of Physiotherapy, University of Melbourne, Australia*
‡ *New England Baptist Hospital, Boston, USA*
§ *Division of Health and Rehabilitation Sciences, University of Queensland, Australia*

ARTICLE INFO

Article history:
Received 23 October 2009
Accepted 5 January 2010

Keywords:
Knee osteoarthritis
Hip strengthening
Joint load
Knee adduction moment

SUMMARY

Objective: To determine whether hip abductor and adductor muscle strengthening reduces medial compartment knee load and improves symptoms in people with medial tibiofemoral OA and varus malalignment.

Methods: In a randomised controlled trial, 89 participants were randomly allocated to a hip strengthening group or to a control group with no intervention. The strengthening group performed a physiotherapist-supervised home exercise program targeting the hip abductor and adductor muscles for 12 weeks. The primary outcome was the peak external knee adduction moment measured using three-dimensional gait analysis by a blinded assessor. Secondary outcomes included a pain numeric rating scale, Western Ontario and McMaster Universities Osteoarthritis Index, step test, stair climb test, maximum isometric strength of hip and quadriceps muscles and participant-perceived rating of overall change. Intention-to-treat analyses were performed using linear regression modelling adjusting for baseline outcomes and other characteristics.

Results: The trial was completed by 76/89 participants (85%). There was no significant between-group difference in change in the knee adduction moment [mean difference (95% confidence interval (CI)) 0.134 (−0.069 to 0.337) Nm/BW × HT%]. All pain, physical function and muscle strength measures showed significantly greater improvement in the strengthening group (all $P < 0.05$). The relative risk (95% CI) of participant-perceived overall improvement in the strengthening group compared to the control group was 20.02 (6.21–64.47).

Conclusions: Although strengthening the hip muscles improved symptoms and function in this patient group, it did not affect medial knee load as measured by the knee adduction moment. Thus it is unlikely that hip muscle strengthening influences structural disease progression. Trial registration: ACTR12607000001493.

© 2010 Osteoarthritis Research Society International. Published by Elsevier Ltd. All rights reserved.

Introduction

Knee osteoarthritis (OA) occurs predominantly in the medial tibiofemoral compartment[1,2] and those with varus knee malalignment are particularly at risk of disease progression[3]. As there is no cure, management for knee OA has targeted symptoms and clinical guidelines reinforce the importance of conservative, non-pharmacologic treatments[4,5]. However, there is increasing interest in identifying treatments that may also slow disease progression. Given that increased knee joint load is strongly implicated in structural deterioration[6], interventions with the potential to both improve symptoms and reduce knee load are worthy of investigation.

As *in vivo* measurement of knee load is difficult, the external knee adduction moment obtained from three-dimensional gait analysis provides a valid and reliable indirect measure of load in the medial tibiofemoral compartment[7–9]. Studies show that patients with knee OA, particularly those with varus malalignment, have a higher peak knee adduction moment when compared to healthy age-matched controls[10,11]. The magnitude of the adduction moment is also predictive of clinical outcomes in knee OA, such as severity of knee pain[12] and radiographic evidence of disease[13]. Importantly, longitudinal data show that as little as a one-unit

* Address correspondence and reprint requests to: Kim L. Bennell, School of Physiotherapy, University of Melbourne, Parkville 3010, Australia. Tel: 61-3-83444135; Fax: 61-3-83444188.

E-mail addresses: k.bennell@unimelb.edu.au (K.L. Bennell), michael.hunt@ubc.ca (M.A. Hunt), timw@unimelb.edu.au (T.V. Wrigley), djhunter@caregroup.harvard.edu (D.J. Hunter), fmcmanus@unimelb.edu.au (F.J. McManus), p.hodges@uq.edu.au (P.W. Hodges), lli7@caregroup.harvard.edu (L. Li), ranash@unimelb.edu.au (R.S. Hinman).

1063-4584/$ – see front matter © 2010 Osteoarthritis Research Society International. Published by Elsevier Ltd. All rights reserved.
doi:10.1016/j.joca.2010.01.010

increase in the adduction moment is associated with up to a 6.5-fold increase in the risk of disease progression[6]. Given the significance of the knee adduction moment with regard to both symptom severity and disease progression in knee OA, conservative strategies to reduce the knee adduction moment constitute a logical rehabilitative approach.

Exercise is recommended for knee OA[14,15] and quadriceps strengthening forms the cornerstone of exercise therapy[16]. Although numerous studies demonstrate the symptomatic benefits of quadriceps strengthening[17–19], we have recently shown that quadriceps strengthening does not influence the knee adduction moment (KAM) in medial knee OA[20]. Furthermore, the only clinical trial evaluating the effect of quadriceps strengthening on structural progression failed to find a significant effect[21]. Taken together, these findings indicate that the role of isolated quadriceps strengthening in the protection against disease progression may be limited.

Muscle weakness in knee OA is not just confined to the quadriceps. Other lower limb muscles including the hip abductors and adductors are also weaker (Hinman *et al.*, unpublished data) and emerging data suggest that these muscles may play a role in disease pathogenesis. Recent evidence implies that hip abductor and adductor muscle strength may be important for reducing the knee adduction moment[22] and a longitudinal cohort study found that people with a lower external *hip* adduction moment (possibly from weaker hip abductor muscles) demonstrated more rapid knee OA progression[23]. The exact mechanism by which hip abductor and adductor muscles may influence medial knee loading is unclear. However, their role has been implicated in stabilising the pelvis and trunk in the frontal plane[22–25] and in eccentric control of frontal plane femoral angles[26]. Thus, hip muscle function appears to be an important, yet under-investigated, contributor to medial knee joint load and has clinical relevance for load-modifying interventions.

The primary aim of this trial was to determine whether strengthening the hip abductor and adductor muscles in people with medial tibiofemoral OA and varus malalignment reduces medial knee joint loading as quantified by the peak external knee adduction moment. The secondary aim was to assess whether a hip strengthening program reduces knee pain and improves physical function.

Methods

Participants

Participants aged over 50 years were recruited from the community through advertisements in newspapers and local clubs and from our database of research volunteers. All had OA in at least one knee fulfilling the American College of Rheumatology classification criteria[27] and reported average knee pain on walking >3 on an 11-point scale (0 = no pain; 10 = maximal pain). To ensure medial tibiofemoral OA and varus malalignment, inclusion criteria were medial knee pain and medial compartment osteophytes or medial joint space narrowing[28] and knee alignment $\leq 182^{\circ}$ on a standardised semiflexed posteroanterior X-ray (corresponding to a mechanical axis of $\leq 180^{\circ}$)[29]. Exclusion criteria were no or doubtful radiographic OA [Kellgren and Lawrence (KL) grades 0 and 1[30]], knee surgery or intra-articular corticosteroid injection within 6 months, current or past (within 4 weeks) oral corticosteroid use, systemic inflammatory arthritic conditions, a history of hip or knee joint replacement or tibial osteotomy, intention to start or currently participating in a supervised lower limb strengthening program, body mass index > 35 due to difficulty in accurate marker placement for gait analysis, a medical condition that precluded safe participation in an exercise program or unable to ambulate without a gait aid.

The trial was performed in compliance with the Helsinki declaration. Ethical approval was obtained from the University of Melbourne Human Research Ethics Committee (HREC no. 0709220). All participants provided written informed consent.

Procedures

We conducted a 12-week, randomised, assessor-blinded controlled trial (Fig. 1). Detailed description of the methodology has been published previously[31]. Potential participants underwent telephone screening, standardised weight bearing X-ray and clinical examination by a physiotherapist to determine eligibility. To include a range of radiographic disease severity, we limited recruitment of participants for each grade of OA severity to 30 (KL grades 2, 3, and 4)[30]. Participants were stratified by radiographic disease severity and randomly allocated in permuted blocks of 4–6 to hip strengthening or control groups according to a computer-generated table of random numbers created by an independent investigator not directly involved in assessment of participants. Allocation was sealed in opaque and consecutively numbered envelopes held in a central location. These were opened in sequence by a person not involved in the study after recruitment and baseline testing of participants.

Interventions

All participants were asked to refrain from seeking other forms of treatment during the trial. However, due to ethical considerations, analgesia and non-steroidal anti-inflammatory drugs were permitted as required.

Hip strengthening

Participants performed a standardised set of home exercises five times per week for 12 weeks[31]. Six exercises to strengthen hip abductor and adductor muscles were performed in side-lying and standing (three sets of 10 repetitions) with ankle cuff weights or elastic bands. Additionally, participants attended a physiotherapy clinic on seven occasions (once weekly for the first 5 weeks then fortnightly) to receive appropriate instruction on exercises and safe progression of resistance. These were individual sessions lasting 30 min initially and 15 min subsequently. The therapists, who had 15.6 (10.1) years of clinical experience and had all been involved in previous research studies, were trained to deliver the exercises prior to the study. They adjusted exercise intensity as determined by the participant's ability to complete 10 repetitions for a given exercise.

Control

Control group participants did not receive any additional intervention or complete any home exercises during the 12 weeks.

Outcome assessment

Participants were assessed at baseline and at 13 weeks by an assessor blinded to group allocation. Age, gender, duration of knee OA symptoms, previous treatment, surgery and medication use for knee OA were obtained at baseline.

Gait measures

Participants underwent three-dimensional gait analyses walking barefoot at a self-selected normal pace along a 10 m walkway with speed monitored by two photoelectric beams. The same speed (±5%) was used at follow-up. Kinematic data were collected using a Vicon motion analysis system with eight M2/MX-F20 CMOS cameras operating at 120 Hz (Vicon, Oxford, UK). The

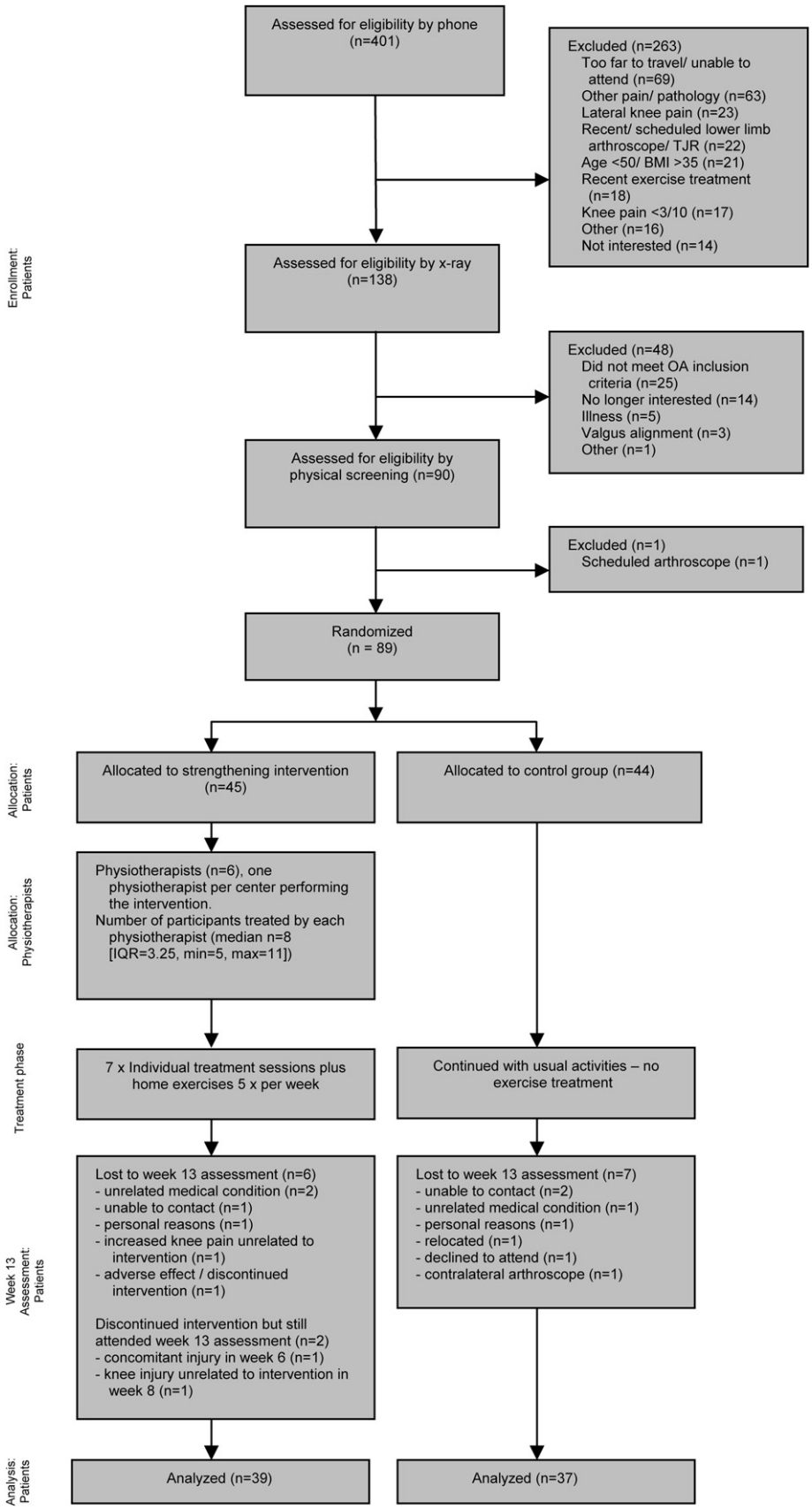

Fig. 1. Trial protocol.

standard Plug-In-Gait marker set was used. Additional markers were placed anteriorly at the sternal notch and posteriorly over the spinous processes of T2 and T10 to construct a local segment coordinate system fixed to the thorax[32] and enable calculation of lateral trunk lean. Lastly, medial knee and ankle markers were included during an initial static standing trial to determine relative positioning of knee and ankle joint centres. Kinetic data were collected using two, floor-mounted 0R6-6-2000 force platforms at a sampling rate of 1080 Hz (Advanced Mechanical Technology Inc., Watertown, MA, USA), in synchrony with the cameras. Net external joint moments were calculated *via* inverse dynamics (Vicon Plug-In-Gait v2). Pain during gait analysis was assessed verbally using an 11-point numerical rating scale (0 = no pain; 10 = maximum pain).

The primary outcome was the peak external KAM during stance phase, normalised to body weight (BW) and height (Nm/BW × HT %). Our test–retest reliability for this measure is excellent [intra-class correlation coefficient ($ICC_{3,5}$) of 0.97]. Other biomechanical variables included the positive area under the KAM curve (knee adduction impulse), peak external hip adduction moment during stance, maximum contralateral pelvic drop, ipsilateral trunk lean in the frontal plane, and foot toe-out angle. Five trials were obtained for the test limb and all discrete data were averaged across trials.

Knee pain and physical function

Overall average knee pain and knee pain on walking in the past week were assessed by 11-point horizontal numeric rating scales with terminal descriptors of 0 = no pain; 10 = maximal pain[33]. Self-reported knee pain and difficulty with physical function were measured using the Western Ontario and McMaster Universities (WOMAC) OA Index[34] where higher scores indicate worse symptoms.

At study completion, participants rated their overall perceived change (compared to baseline) on an ordinal scale (1 – much worse, 2 – slightly worse, 3 – no change, 4 – slightly better, 5 – much better)[35]. Improvement was deemed *a priori* as 'slightly better' or 'much better'.

Objective physical function tests included the step test and a timed stair ascent/descent task. The step test is a functional, dynamic test of standing balance[36] whereby participants stand barefoot on the study leg in front of a 15 cm step and step the opposite foot on and off the step as quickly as possible over 15 s. The number of times the foot can be placed up onto the step and returned to the floor is recorded, with higher scores suggesting better balance. The stair ascent/descent task involves the timed ascent and descent of six stairs, with longer times indicating poorer physical function[37].

Strength measures

Maximal isometric strength of hip abduction and hip adduction were measured in supine while hip flexion, internal rotation and external rotation were measured in sitting using a handheld dynamometer (Nicholas MMT, Lafayette Instruments, Lafayette, IN, USA)[38]. Hip extension strength was measured in supine using a force transducer[38]. Isometric quadriceps muscle strength at 60° of knee flexion was assessed in sitting using the Kin-Com 125-AP dynamometer (Chattecx Corporation, Chattanooga, TN, USA)[20].

After a single, submaximal trial, participants performed three trials each of 5 s duration, separated by 15 s of rest. The maximum force output (*N*) from the three trials (corrected for gravitational weight of the limb as appropriate) was recorded and converted to torque (Nm) by multiplying by the resistance lever arm (*m*), then normalised to body mass (Nm/kg). These strength tests have excellent test–retest reliability in our laboratory (ICCs = 0.84–0.98).

Other measures

Physical activity was measured using the physical activity scale for the elderly (PASE) which assesses the level and type of recreational and occupational physical activities undertaken by participants over the previous week[39].

Co-interventions, medication use and adverse events were recorded by participants in a weekly logbook. Adherence in the strengthening group was assessed as the number of physiotherapy sessions attended (maximum of seven converted to a percentage) and the number of home exercise sessions performed (maximum of 60 exercise days converted to a percentage).

Sample size calculations

We estimated an achievable and clinically relevant change in the peak knee adduction moment as being 7.5% because this may be associated with an approximate 2-fold reduction in the risk of structural disease progression[6,40] and because other non-pharmacological conservative interventions such as lateral wedge insoles can achieve reductions of this magnitude[6,40]. Based on our previous studies, a 7.5% reduction equates to an absolute change of approximately 0.3 Nm/BW × HT%[20,40]. Estimates of the standard deviation of change are 0.4 Nm/BW × HT% based on our data and

Table I
Baseline characteristics of total participants and of those who were lost to follow-up in both groups

Characteristic	Participants			
	Randomised ($n = 89$)		Lost to follow-up ($n = 13$)	
	Exercise ($n = 45$)	Control ($n = 44$)	Exercise ($n = 6$)	Control ($n = 7$)
Age (yr), mean (SD)	64.5 (9.1)	64.6 (7.6)	62.7 (10.3)	62.7 (8.2)
Gender, *n* males (%)	22 (48.9%)	24 (54.5%)	2 (33.3%)	3 (42.9%)
Knee affected, *n* right (%)	27 (60.0%)	21 (47.7%)	3 (50.0%)	3 (42.9%)
Symptoms, *n* unilateral (%)	18 (40.0%)	15 (34.1%)	1 (16.7%)	2 (28.6%)
Body mass index (kg/m^2), mean (SD)	27.5 (4.7)	28.4 (4.1)	28.7 (4.0)	30.2 (3.7)
Static knee alignment (°), mean (SD)	176.4 (2.6)	177.2 (2.3)	175.3 (4.7)	177.6 (2.1)
KL grade n (%)				
Grade 2	15 (33.3%)	15 (34.1%)	1 (16.7%)	3 (42.9%)
Grade 3	15 (33.3%)	14 (31.8%)	2 (33.3%)	0 (0%)
Grade 4	15 (33.3%)	15 (34.1%)	3 (50.0%)	4 (57.1%)
Physical activity levels, mean (SD)†	251.1 (88.1)	237.6 (81.0)	211.6 (91.1)	240.7 (56.7)
Current medication use *n* (%)	28 (62.2%)	33 (75.0%)	4 (66.7%)	6 (85.7%)
Analgesics	7 (15.6%)	14 (31.8%)	0 (0.0%)	2 (28.6%)
Non-steroidal anti-inflammatories	11 (24.4%)	11 (25.0%)	2 (33.3%)	3 (42.9%)
Nutraceuticals (e.g., glucosamine)	17 (37.8%)	20 (45.5%)	3 (50.0%)	2 (28.6%)

† Measured using the PASE from 0 to 400 with higher scores indicating higher activity levels.

Table II
Mean (SD) of groups, mean (SE) difference within groups, and mean (95% CI) difference between groups for the primary outcome and other biomechanical variables

	Groups, mean (SD)				Difference within groups, adjusted mean (SE)†		Difference between groups, adjusted mean (95% CI)	
	Week 0		Week 13		Week 13–Week 0		Week 13–Week 0	
	Exercise	Control	Exercise	Control	Exercise	Control	Exercise–Control	*P*-values
Peak knee adduction moment (Nm/BW × HT%)	3.2 (1.0)	2.9 (0.9)	3.3 (0.9)	2.9 (0.9)	0.15 (0.07)	0.02 (0.07)	0.13 (−0.07, 0.34)	0.193
Peak hip adduction moment (Nm/BW × HT%)	5.3 (0.9)	5.1 (1.2)	5.4 (0.9)	5.2 (1.1)	0.12 (0.11)	0.04 (0.11)	0.08 (−0.23, 0.39)	0.620
Knee adduction impulse	1.1 (0.4)	1.0 (0.4)	1.1 (0.4)	1.0 (0.4)	0.05 (0.03)	0.02 (0.03)	0.04 (−0.05, 0.12)	0.407
Maximum contralateral pelvic drop (deg)	3.7 (2.6)	3.0 (2.2)	4.0 (2.5)	2.9 (2.0)	0.54 (0.21)	−0.22 (0.21)	0.77 (0.15, 1.38)	0.015
Maximum ipsilateral trunk lean (deg)	3.3 (2.2)	4.0 (2.7)	2.7 (1.8)	3.7 (2.7)	−0.74 (0.23)	−0.20 (0.23)	−0.54 (−1.21, 0.14)	0.116
Toe-out in stance (deg)	6.0 (4.6)	7.3 (5.4)	5.6 (4.6)	5.9 (5.2)	−0.37 (0.35)	−1.38 (0.36)	1.01 (−0.02, 2.03)	0.055

† Peak knee adduction moment adjusted for baseline value, KL grade, change in pain during gait testing and change in walking speed. Other biomechanical variables were adjusted for baseline value, KL grade and change in pain during gait testing.

others[41]. Thirty-eight participants per group were required to detect a 0.3 Nm/BW × HT% change with 90% power (alpha = 0.05). We assumed an attrition rate of 15%, thus at least 44 participants per group were required.

Statistical analysis

Analyses were performed blinded and using an intention-to-treat principle. Missing data were replaced using the last observation carried forward. For continuous outcomes, differences in mean change between groups were evaluated using linear regression modelling with adjustment for baseline scores. For the knee adduction moment, additional covariates included change in gait speed, disease severity, and change in walking pain during testing. For other biomechanical variables, additional covariates were disease severity and change in walking pain during testing. These covariates were chosen given some evidence of their association with the knee adduction moment (e.g., Ref. 12). Model assumptions were checked by standard diagnostic plots. Participant rating of overall perceived change in pain and function was compared between groups by calculating relative risks with 95% confidence intervals using log binomial regression.

Results

Eighty-nine participants were recruited (45 strengthening, 44 control) between October 2007–January 2009 and 76 (39 strengthening, 37 control; 85%) completed the trial (Fig. 1). Participant characteristics are shown in Table I, with groups similar at baseline. Characteristics of participants lost to follow-up did not differ to those completing the study (Table I). Assessor unblinding occurred in six participants in the exercise group, three of these during follow-up testing (participants inadvertently mentioning exercises), and three immediately after follow-up data collection (participants questioned whether to continue exercises).

Knee loading and biomechanics

Table II demonstrates no difference between strengthening and control groups for change in the external knee adduction moment [0.148 (95% confidence interval (CI) −0.039 to 0.335) Nm/BW × HT %]. In fact, rather than reduced knee load as hypothesised, the strengthening group showed a tendency towards (4.6% adjusted) increased knee adduction moment over the 12 weeks [compared with 0.5% adjusted increase in control participants ($P = 0.19$)]. Results were unchanged when change in walking pain during gait analysis was removed as a covariate ($P = 0.12$) or when only individuals who completed the study were analysed ($P = 0.11$).

There were no between-group differences in change of peak external hip adduction moment, knee adduction impulse, or maximum ipsilateral trunk lean. Significant between-group group differences were noted for change in maximum contralateral pelvic drop [0.765 (95% CI 0.152–1.379)°] whereby the strengthening group showed a 15% increase and the controls a 7% decrease ($P = 0.015$).

Pain and function

All measures of pain and physical function showed significantly greater improvements in the strengthening group compared with the control group (all $P < 0.05$) (Table III). Improved pain and physical function was reported by 31/39 (80%) participants in the

Table III
Mean (SD) of groups, mean (SE) difference within groups, and mean (95% CI) difference between groups for pain, physical function and strength

	Groups, mean (SD)				Difference within groups, adjusted mean (SE)†		Difference between groups, adjusted mean (95% CI)†	
	Week 0		Week 13		Week 13–Week 0		Week 13–Week 0	
	Exercise	Control	Exercise	Control	Exercise	Control	Exercise–Control	*P*-values
Pain on walking (0–10)	4.3 (2.0)	4.1 (2.5)	2.6 (2.1)	3.9 (2.6)	−1.61 (0.28)	−0.24 (0.28)	−1.37 (−2.16, −0.59)	0.0008
WOMAC function (0–68)	24.8 (10.9)	23.7 (11.8)	16.2 (11.7)	21.9 (11.0)	−8.07 (1.15)	−1.90 (1.16)	−6.17 (−9.41, −2.93)	0.0003
WOMAC pain (0–20)	7.7 (3.0)	6.9 (3.3)	4.9 (3.3)	6.5 (3.3)	−2.60 (0.39)	−0.48 (0.40)	−2.12 (−3.24, −1.00)	0.0003
Step test (*n*)	16.2 (3.7)	16 (2.9)	18 (4.3)	16.9 (2.8)	1.76 (0.32)	0.80 (0.32)	0.96 (0.07, 1.86)	0.0358
Timed stair task (s)	8.0 (2.7)	8.2 (2.3)	7.0 (2.2)	7.9 (1.8)	−0.97 (0.19)	−0.25 (0.19)	−0.73 (−1.25, −0.20)	0.0072
Knee extension torque (Nm/kg)	1.5 (0.5)	1.4 (0.6)	1.6 (0.4)	1.4 (0.5)	0.16 (0.03)	0.05 (0.03)	0.11 (0.02, 0.21)	0.0152
Hip flexion torque (Nm/kg)	0.8 (0.2)	0.8 (0.2)	0.8 (0.2)	0.7 (0.2)	0.07 (0.03)	−0.01 (0.03)	0.08 (0.01, 0.16)	0.0325
Hip abduction torque (Nm/kg)	0.9 (0.3)	0.8 (0.3)	1.0 (0.3)	0.9 (0.3)	0.14 (0.03)	0.03 (0.03)	0.11 (0.04, 0.19)	0.0022
Hip adduction torque (Nm/kg)	0.8 (0.3)	0.8 (0.3)	0.9 (0.3)	0.8 (0.3)	0.15 (0.02)	0.024 (0.023)	0.13 (0.06, 0.19)	0.0002
Hip extension torque (Nm/kg)	1.9 (0.6)	1.8 (0.6)	2.1 (0.7)	1.9 (0.6)	0.19 (0.04)	0.04 (0.04)	0.15 (0.04, 0.27)	0.0115
Hip internal rotation torque (Nm/kg)	0.4 (0.2)	0.4 (0.2)	0.5 (0.2)	0.4 (0.2)	0.04 (0.02)	−0.01 (0.02)	0.05 (0.01, 0.10)	0.0269
Hip external rotation torque (Nm/kg)	0.4 (0.1)	0.4 (0.2)	0.4 (0.1)	0.4 (0.2)	0.04 (0.01)	−0.02 (0.01)	0.05 (0.02, 0.09)	0.0048

† Values were adjusted for the baseline value of the variable.

strengthening group but by only 6/37 (16%) in the control group ($P < 0.0001$). The relative risk (95% CI) of improvement in the strengthening group compared to the control group was 20.02 (6.21–64.47).

Strength

Changes in hip strength were significantly different between groups (Table III). In particular, the hip abductor and adductor strength increased by 13% and 19%, respectively, in the strengthening group, compared with 4% and 3%, respectively, in the control group. The strengthening group also showed a significantly greater increase in knee extension strength compared to controls (11% vs 3% $P = 0.015$).

Adherence, co-interventions and adverse effects

For the strengthening group, the mean (SD) physiotherapy attendance rate and home exercise compliance rate were 96% (11%) and 89% (7%), respectively. Five participants reported an adverse event: three reported back pain, one reported back and hip pain, and one reported aggravated varicose veins and knee pain for which acupuncture treatment was sought. Only one participant in the strengthening group (in addition to the one requiring acupuncture noted above) sought a co-intervention (ultrasound and strapping from a physiotherapist). In the control group, seven participants sought treatment including knee arthroscopy ($n = 1$), physiotherapy and massage ($n = 1$), osteopathy ($n = 1$), trigger point needling and non-steroidal anti-inflammatory gel ($n = 1$), massage ($n = 1$), and topical creams ($n = 2$). Eighteen exercise participants and 14 controls reported increased medication use, whilst 15 and nine, respectively reported decreased medication use at any point in the trial.

Discussion

This study showed that hip strengthening had no significant effect on the external knee adduction moment, an indicator of medial tibiofemoral compartment load during walking and a biomechanical factor linked to disease progression, in people with medial tibiofemoral OA and varus malalignment. Despite no effect on the knee adduction moment, the hip strengthening program did lead to significant improvements in pain and physical function.

This is the first randomised controlled trial to investigate the specific effects of hip muscle strengthening in knee OA. Our research extends the work of others in this area. An 18-month longitudinal observational study indicated that a lower external hip adduction moment (internal hip abduction moment) was associated with an increased risk of medial knee OA structural progression[23]. To explain their findings, Chang *et al.*[23] proposed that weaker hip abductors – assumed to be associated with a *lower* external hip adduction moment – on the osteoarthritic stance limb would result in additional pelvic drop of the contralateral swing limb, shifting the body's centre of mass towards the swing limb. This would lengthen the lever arm at the osteoarthritic knee, thus increasing medial knee load and hastening disease progression. This hypothesis is supported by a cross-sectional study that showed that those with more severe knee OA had lower external hip adduction moments and higher knee adduction moments than those with less severe disease[22]. However, although our exercise program achieved a 13% increase in hip abductor strength, we observed no change in the external hip adduction moment, a significant *increase* in contralateral pelvic drop, and no change in the external knee adduction moment.

Although it is often presumed that the external hip adduction moment reflects hip abductor strength, this is not necessarily the case as the relationship is more complex. This moment, measured by inverse dynamics, is a *net* moment and cannot be attributed to any single muscle group; rather it represents the overall effect of all moment-generators (both active and passive structures) in both abduction and adduction directions. Furthermore, the moment will be influenced by movement characteristics, particularly at the pelvis and trunk, as these will determine the extent to which hip muscles are required to be activated. As such, there may not necessarily be a close relationship between hip abductor strength and the hip adduction moment as demonstrated recently in a study of healthy individuals[42]. Furthermore, Henriksen *et al.*[43] found that a reduction in the external hip adduction moment associated with experimental pain-induced hip abductor weakness in healthy individuals led to a reduction, rather than an increase, in the knee adduction moment. This further challenges the premise that hip muscle strengthening would lead to a reduction in the knee adduction moment. Taken together with our results, these disparate findings highlight the complexity of the relationship between hip muscle strength, muscle activation levels, hip and pelvic movement patterns, hip moments and ultimately medial knee load.

It might be argued that our failure to demonstrate changed external hip and knee adduction moments was because our cohort did not exhibit substantial hip muscle weakness in the first place. However, comparison of our cohort with an age-matched asymptomatic control group ($n = 24$) revealed adjusted average strength deficits in the order of 23% for the hip abductors and 25% for the hip adductors[49]. Instead, it may be that our patients altered their gait mechanics following hip strengthening in such a manner that counteracted any potential benefits for knee loading. For example, the exercise group exhibited a small but significant decrease in lateral trunk lean (likely associated with the increase in pelvic drop) with the hip strengthening program, consistent with improvements in hip abductor strength and reductions in pain. Reduced trunk lean is known to *increase* the knee adduction moment[24,25]. Thus, it is possible that any direct effects of improved hip strength on the knee adduction moment may have been cancelled out by reductions in compensatory lateral trunk lean. It is also possible that our participants did not use their increased hip muscle strength, despite the overall strength increases we observed. Although our training program provided participants with greater potential to advantageously use hip muscle strength, it did not specifically teach participants how to use their enhanced strength functionally. Perhaps a motor control program incorporating eccentric hip abductor training in single leg stance and co-ordination during functional activities have been more specific to achieving a reduction in the external knee adduction moment.

Relative to hip abductor function, less is known about the hip adductor muscles in knee OA. By virtue of their attachment to the distal medial femoral condyle, the adductors could eccentrically restrain the tendency of the femur to move further into varus thus reducing the knee adduction moment. Yamada *et al.*[26] found that patients with knee OA had stronger hip adductors compared with age-matched controls, and that those with more severe OA had even stronger adductors than their less severe counterparts. They hypothesised that this increased strength may be due to greater eccentric use of the hip adductors to control femoral varus and lower the knee adduction moment. We anticipated that addition of hip adductor strengthening would complement the proposed benefits of improved hip abductor strength on medial knee load. However, our results do not lend support to a beneficial effect of a combined program including hip adductor strengthening on the knee adduction moment.

Clinically important improvements in pain and physical function were found in the hip strengthening group. It is possible that some symptomatic benefit of strengthening may be attributed to indirect (placebo) treatment effects related to the therapeutic environment and/or expectation of benefit[44]. Given that our control group received no treatment rather than placebo treatment, this cannot be ascertained. However, given that placebo effects are typically confined to pain and other self-report measures[45], our improvements in objective measures of physical function are unlikely to be due simply to a placebo effect but rather to the direct effects of the exercises.

The clinical improvements noted are consistent with the results of numerous studies demonstrating a beneficial effect of strengthening on pain and function in knee OA[16]. While our exercises specifically targeted the hip musculature, the quadriceps also improved in strength given their need to maintain knee extension during many of the hip exercises. As muscle strength, particularly of the quadriceps, is an important predictor of both pain and physical function[46,47], it is not surprising that improvements in muscle strength translated into symptomatic improvement. Regardless of the mechanism of effect, our results indicate that a hip strengthening program can provide symptomatic benefits for patients with knee OA. This may be particularly useful for patients in whom quadriceps exercises are too painful.

The strengths of our study include the rigorous study design, specific recruitment of patients with varus malalignment, excellent participant retention and adherence, and successful strengthening of the target muscle groups. A limitation is that the participants and therapists were not blinded to the intervention, which is difficult in a trial of this nature. However, this is unlikely to have biased the primary outcome given the use of an objective measurement of the knee adduction moment. As acknowledged, the lack of an intervention for the control group may have influenced the outcomes for the secondary self-report measures of pain and function. Furthermore, we did not directly assess structural disease progression. Rather, we used the commonly accepted surrogate measure of medial compartment load – the external knee adduction moment which has been related to disease progression[6]. However, it is conceivable that participants experienced changes in medial compartment loading that were not reflected in changes in the external adduction moment. Thus, while our results do not support hip strengthening as an OA disease-modifying intervention, this would need to be confirmed using direct measures of structural change such as magnetic resonance imaging. Another issue worthy of further consideration is whether specific patient characteristics predict a better response to hip strengthening treatment as this may highlight patient subgroups in whom the treatment might be most effective for improving symptoms[48].

In summary, our data do not support hip muscle strengthening as a potentially disease-modifying intervention in medial knee OA. We found that strengthening the hip abductor and adductor muscles improved symptoms in this patient group, but did not affect medial knee load as measured by the knee adduction moment. Hence the mechanism by which a higher hip adduction moment is protective against disease progression[23] is unclear. Further work is warranted to investigate if other conservative interventions can increase the hip adduction moment, reduce medial knee load and slow structural disease progression.

Conflict of interest

The authors declare that they have no conflicts of interest.

Acknowledgements

This trial was funded by the National Health and Medical Research Council (Project #454686).

Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi: 10.1016/j.joca.2010.01.010.

References

1. Ledingham J, Regan M, Jones A, Doherty M. Radiographic patterns and associations of osteoarthritis of the knee in patients referred to hospital. Ann Rheum Dis 1993;52:520–6.
2. Iorio R, Healy W. Unicompartmental arthritis of the knee. J Bone Joint Surg 2003;85-A:1351–64.
3. Tanamas S, Hanna FS, Cicuttini FM, Wluka AE, Berry P, Urquhart DM. Does knee malalignment increase the risk of development and progression of knee osteoarthritis? A systematic review. Arthritis Rheum 2009;61:459–67.
4. Jordan K, Arden N, Doherty M, Bannwarth B, Bijlsma J, Dieppe P, *et al*. EULAR recommendations 2003: an evidence based approach to the management of knee osteoarthritis: report of a task force of the Standing Committee for International Clinical Studies Including Therapeutic Trials (ESCISIT). Ann Rheum Dis 2003;62:1145–55.
5. ACR Subcommittee on OA. Recommendations for the medical management of osteoarthritis of the hip and knee. 2000 update. Arthritis Rheum 2000;43:1905–15.
6. Miyazaki T, Wada M, Kawahara H, Sato M, Baba H, Shimada S. Dynamic load at baseline can predict radiographic disease progression in medial compartment knee osteoarthritis. Ann Rheum Dis 2002;61:617–22.
7. Birmingham TB, Hunt MA, Jones IC, Jenkyn TR, Giffin JR. Test–retest reliability of the peak knee adduction moment during walking in patients with medial compartment knee osteoarthritis. Arthritis Rheum 2007;57:1012–7.
8. Schipplein OD, Andriacchi TP. Interaction between active and passive knee stabilizers during level walking. J Orthop Res 1991;9:113–9.
9. Zhao D, Banks SA, Mitchell KH, D'Lima DD, Colwell Jr CW, Fregly BJ. Correlation between the knee adduction torque and medial contact force for a variety of gait patterns. J Orthop Res 2007;25:789–97.
10. Baliunas A, Hurwitz D, Ryals A, Karrar A, Case J, Block J, *et al*. Increased knee joint loads during walking are present in subjects with knee osteoarthritis. Osteoarthritis Cartilage 2002;10:573–9.
11. Hurwitz D, Ryals A, Case J, Block J, Andriacchi T. The knee adduction moment during gait in subjects with knee osteoarthritis is more closely correlated with static alignment than radiographic disease severity, toe out angle and pain. J Orthop Res 2002;20:101–8.
12. Shrader M, Draganich L, Pottenger L, Piotrowski G. Effects of knee pain relief in osteoarthritis on gait and stair-stepping. Clin Orthop Relat Res 2004;421:188–93.
13. Sharma L, Hurwitz DE, Thonar E, Sum JA, Lenz ME, Dunlop DD, *et al*. Knee adduction moment, serum hyaluronan level, and disease severity in medial tibiofemoral osteoarthritis. Arthritis Rheum 1998;41:1233–40.
14. Roddy E, Doherty M. Guidelines for management of osteoarthritis published by the American College of Rheumatology and the European League Against Rheumatism: why are they so different? Rheum Dis Clin N Am 2003;29:717–31.
15. Zhang W, Moskowitz RW, Nuki G, Abramson S, Altman RD, Arden N, *et al*. OARSI recommendations for the management of hip and knee osteoarthritis, part II: OARSI evidence-based, expert consensus guidelines. Osteoarthritis Cartilage 2008;16:137–62.

16. Fransen M, McConnell S. Land-based exercise for osteoarthritis of the knee: a metaanalysis of randomized controlled trials. J Rheumatol 2009;36:1109–17.
17. Fransen M, Crosbie J, Edmonds J. Physical therapy is effective for patients with osteoarthritis of the knee: a randomized controlled trial. J Rheumatol 2001;28:156–64.
18. Huang M-H, Lin Y-S, Yang R-C, Lee C-L. A comparison of various therapeutic exercises on the functional status of patients with knee osteoarthritis. Semin Arthritis Rheum 2003;32:398–406.
19. Mikesky AE, Meyer A, Thompson KL. Relationship between quadriceps strength and rate of loading during gait in women. J Orthop Res 2000;18:171–5.
20. Lim BW, Hinman RS, Wrigley TV, Sharma L, Bennell KL. Does knee malalignment mediate the effects of quadriceps strengthening on knee adduction moment, pain, and function in medial knee osteoarthritis? A randomized controlled trial. Arthritis Rheum 2008;59:943–51.
21. Mikesky A, Mazzuca S, Brandt K, Perkins S, Damush T, Lane K. Effects of strength training on the incidence and progression of knee osteoarthritis. Arthritis Rheum 2006;55:690–9.
22. Mundermann A, Dyrby C, Andriacchi T. Secondary gait changes in patients with medial compartment knee osteoarthritis: increased load at the ankle, knee, and hip during walking. Arthritis Rheum 2005;52:2835–44.
23. Chang A, Hayes K, Dunlop D, Song J, Hurwitz D, Cahue S, *et al*. Hip abduction moment and protection against medial tibiofemoral osteoarthritis progression. Arthritis Rheum 2005;52:3515–9.
24. Hunt MA, Birmingham TB, Bryant D, Jones I, Giffin JR, Jenkyn TR, *et al*. Lateral trunk lean explains variation in dynamic knee joint load in patients with medial compartment knee osteoarthritis. Osteoarthritis Cartilage 2008;16:591–9.
25. Mundermann A, Asay JL, Mundermann L, Andriacchi TP. Implications of increased medio-lateral trunk sway for ambulatory mechanics. J Biomech 2008;41:165–70.
26. Yamada H, Koshino T, Sakai N, Saito T. Hip adductor muscle strength in patients with varus deformed knee. Clin Orthop Relat Res 2001;386:179–85.
27. Altman R, Asch E, Bloch D, Bole G, Borenstein D, Brandt K, *et al*. Development of criteria for the classification and reporting of osteoarthritis. Classification of osteoarthritis of the knee. Arthritis Rheum 1986;29:1039–49.
28. Altman R. The classification of osteoarthritis. J Rheumatol 1995;22:42–3.
29. Kraus V, Parker Vail T, Worrell T, McDaniel G. A comparative assessment of alignment angle of the knee by radiographic and physical examination methods. Arthritis Rheum 2005;52:1730–5.
30. Kellgren JH, Lawrence JS. Radiological assessment of osteoarthrosis. Ann Rheum Dis 1957;16:494–501.
31. Bennell KL, Hunt MA, Wrigley TV, Hunter DJ, Hinman RS. The effects of hip muscle strengthening on knee load, pain, and function in people with knee osteoarthritis: a protocol for a randomised, single-blind controlled trial. BMC Musculoskelet Disord 2007;8:121.
32. Baker R. Modelling concepts in motion analysis. In: JEGM, Amsterdam, Netherlands 2006.
33. Bellamy N. Osteoarthritis clinical trials: candidate variables and clinimetric properties. J Rheumatol 1997;24:768–78.
34. Bellamy N, Kirwan J, Boers M, Brooks P, Strand V, Tugwell P, *et al*. Recommendations for a core set of outcome measures for future phase iii clinical trials in knee, hip, and hand osteoarthritis – consensus development at OMERACT III. J Rheumatol 1997;24:799–802.
35. Crossley KM, Bennell KL, Cowan SM, Green S. Analysis of outcome measures for persons with patellofemoral pain: which are reliable and valid? Arch Phys Med Rehabil 2004;85: 815–22.
36. Hill K, Bernhardt J, McGann AM, Maltese D, Berkovits D. A new test of dynamic standing balance for stroke patients: reliability, validity and comparison with healthy elderly. Physiother Can 1996;48:257–62.
37. Rejeski WJ, Ettinger Jr WH, Schumaker S, James P, Burns R, Elam JT. Assessing performance-related disability in patients with knee osteoarthritis. Osteoarthritis Cartilage 1995;3:157–67.
38. Pua YH, Wrigley TV, Cowan SM, Bennell KL. Intrarater test–retest reliability of hip range of motion and hip muscle strength measurements in persons with hip osteoarthritis. Arch Phys Med Rehabil 2008;89:1146–54.
39. Washburn R, Smith K, Jette A, Janney CA. The physical activity scale for the elderly (PASE): development and evaluation. J Clin Epidemiol 1993;46:153–62.
40. Hinman RS, Bowles KA, Payne C, Bennell KL. Effect of length on laterally-wedged insoles in knee osteoarthritis. Arthritis Rheum 2008;59:144–7.
41. Kerrigan D, Lelas J, Goggins J, Merriman G, Kaplan R, Felson D. Effectiveness of a lateral-wedge insole on knee varus torque in patients with knee osteoarthritis. Arch Phys Med Rehabil 2002;83:889–93.
42. Rutherford DJ, Hubley-Kozey C. Explaining the hip adduction moment variability during gait: implications for hip abductor strengthening. Clin Biomech 2009;24:267–73.
43. Henriksen M, Aaboe J, Simonsen EB, Alkjaer T, Bliddal H. Experimentally reduced hip abductor function during walking: implications for knee joint loads. J Biomech 2009;42:1236–40.
44. Crow R, Gage H, Hampson S, Hart J, Kimber A, Thomas H. The role of expectancies in the placebo effect and their use in the delivery of health care: a systematic review. Health Technol Assess 1999;3:1–96.
45. Hrobjartsson A, Gotzsche P. Is the placebo powerless? An analysis of clinical trials comparing placebo with no treatment. N Engl J Med 2001;344:1594–602.
46. Maly M, Costigan P, Olney S. Determinants of self-report outcome measures in people with knee osteoarthritis. Arch Phys Med Rehabil 2006;87:96–104.
47. van Dijk GM, Veenhof C, Lankhorst GJ, Dekker J. Limitations in activities in patients with osteoarthritis of the hip or knee: the relationship with body functions, comorbidity and cognitive functioning. Disabil Rehabil 2009;1–7.
48. Hancock M, Herbert RD, Maher CG. A guide to interpretation of studies investigating subgroups of responders to physical therapy interventions. Phys Ther 2009;89:698–704.
49. Hinman RS, Hunt MA, Creaby MW, Wrigley TV, McManus FJ, Bennell KL. Hip muscle weakness is present in individuals with medial knee osteoarthritis. Arthritis Rheum (in press).